LAP TIME SIMULATOR (laptimesim)

High-Fidelity 18+ DOF Multi-Body Dynamics & Optimal Control Simulation Report

1. Executive Summary

This document presents the detailed architectural formulation, mathematical modeling, and complete parameter specification of the **laptimesim** vehicle dynamics simulation package. The software was built for Formula Student electric racing teams and reduced-order modeling (ROM) research. The solver formulates a minimum-time **Optimal Control Problem (OCP)** in curvilinear (Frenet) coordinates along the circuit centerline, transcribed via direct collocation into a large-scale nonlinear programming problem (NLP) solved with **CasADi and IPOPT (MUMPS)**.

A full-vehicle benchmark was executed on the official **Formula Student Germany (FSG at Hockenheimring) Endurance Circuit (903 m)**. The vehicle model incorporates full multi-body suspension load transfers, Pacejka combined slip tire interactions, multi-dimensional aerodynamic ride-height maps, and electric powertrain power limits.

2. Mathematical Formulation & Degrees of Freedom

The simulation models the vehicle with **18+ continuous physical degrees of freedom**, described by the state vector **x(s)** and control vector **u(s)** where arc length *s* is the independent variable.

Subsystem	States / DOFs	Physics Governed
Chassis 6-DOF	vx, vy, vz, phi, theta, psi_dot, phi_dot, theta_dot	Coupled Newton-Euler equations of rigid body translation and rotation in 3D space.
Wheel Spin (4x)	omega_FL, omega_FR, omega_RL, omega_RR	Rotational inertia dynamics driven by motor/brake torques and longitudinal tire traction.
Suspension Travel (4x)	zs_FL, zs_FR, zs_RL, zs_RR, dzs_FL, dzs_FR, dzs_RL, dzs_RR	Quarter-car deflections, damping forces, anti-roll bars (ARB), and progressive elastomeric bump stops.
Compliances	delta_compl, theta_dt	Steering rack/column torsional compliance and rear drivetrain axle torsional twist.
Battery Energy & Thermal	soc, t_batt	Equivalent Circuit Model (ECM Rint) tracking State of Charge, Joule heating, and terminal voltage.
Curvilinear Tracking	n, psi	Lateral offset from centerline n(s) and heading error angle relative to track tangent psi(s).

Curvilinear Spatial Transformation:

To formulate the objective strictly as minimum time, the problem transforms time-derivatives to spatial derivatives via the progress speed equation:

$$\frac{dt}{ds} = \frac{1 - n\cdot\kappa(s)}{vx\cdot\cos(\psi) - vy\cdot\sin(\psi)}$$

$$\frac{dx_{veh}}{ds} = \frac{dx_{veh}}{dt} \cdot \frac{dt}{ds}$$

$$\text{Objective: } \min \int \frac{dt}{ds}\, ds$$

3. Detailed Parameter Tables & Maps

The following parameters define the vehicle configuration used in the benchmark simulation (matching *examples/fsg_benchmark.toml*):

Subsystem	Parameter	Value	Unit / Description
Chassis & Inertia	Total Mass (m)	290.0	kg (215 kg car + 75 kg driver)
	Wheelbase (L)	1.530	m
	Track Width Front / Rear	1.200 / 1.180	m
	CG Height (h_cg)	0.280	m
	Weight Distribution (Front / Rear)	48% / 52%	Longitudinal CG balance
	Inertia Tensor (Ixx, Iyy, Izz)	78.0, 110.0, 90.0	kg·m² (Roll, Pitch, Yaw)
Pacejka MF 5.2 Tire	Stiffness Factor (B)	Front: 10.5 | Rear: 11.0	TTC Hoosier 18.0x7.5-10 fitment
	Shape Factor (C)	Front: 1.65 | Rear: 1.68	Peak sharpness
	Peak Grip Factor (D)	Front: 1.55 | Rear: 1.60	Peak friction coefficient (μ)
	Curvature Factor (E)	-0.40	Post-peak falloff profile
	Loaded Radius (r_wheel)	0.2032 (8.0 in)	m
Aerodynamics	Lift Coefficient (Cl)	3.60	High-downforce multi-element wings
	Drag Coefficient (Cd)	1.45	Aerodynamic drag factor
	Frontal Area (A)	1.15	m²
	Center of Pressure (CoP)	46%	Aero downforce front distribution
Powertrain (EV)	Peak Motor Power	80.0	kW (FSG rule ceiling)
	Peak Motor Torque	240.0	Nm
	Gearbox Reduction Ratio	3.60	Single-speed planetary / spur
	Drivetrain Efficiency (η)	94.0%	Inverter + gearbox transmission
	Regen Braking Torque Cap	60.0	Nm (Front/Rear regenerative recovery)
Battery Pack ECM	Capacity / Voltage	6.8 kWh / 400 V	Nominal pack energy & voltage
	Internal Resistance (R_int)	0.045	Ohms (total accumulator)
Suspension	Spring Rates (Front / Rear)	38,000 / 42,000	N/m (Linear main springs)
	Damper Rates (Front / Rear)	1,900 / 2,100	N·s/m (Damping coefficients)
	ARB Stiffness (Front / Rear)	5,500 / 4,500	N/m (Anti-Roll Bar roll resistance)

4. Aerodynamic & Nonlinear Look-up Maps

Multi-Dimensional Aerodynamic Ride Height Map (B-spline Surface):
In the extended aero module (*AeroMapModel*), Cl, Cd, and balance are functions of front and rear ride heights:
• Grid X (Front Ride Height): [20 mm, 30 mm, 40 mm, 50 mm]
• Grid Y (Rear Ride Height): [25 mm, 35 mm, 45 mm, 55 mm]
• Cl Map Matrix: [[3.4, 3.3, 3.1, 2.9], [3.5, 3.4, 3.2, 3.0], [3.3, 3.2, 3.0, 2.8], [3.1, 3.0, 2.8, 2.6]]
• Front Aero Balance: Evaluated using continuous 2D B-spline interpolation in CasADi ensuring C^2 differentiability for IPOPT.

Nonlinear Suspension & Bump Stop Curve:
When suspension travel *z_s* exceeds bump clearance (15 mm front, 18 mm rear), a progressive elastomeric force is activated:

F_bump = k_bump · (max(0, z_s - clearance))^1.5

where *k_bump* = 120,000 N/m (front) and 140,000 N/m (rear), preventing chassis grounding under high aerodynamic downforce.

5. Simulation Results & Performance Telemetry

The optimal trajectory was computed over the **903 m FSG Endurance circuit**. Key performance telemetry metrics include:

Simulation Metric	Calculated Result	Significance / Competitiveness
Endurance Lap Time	39.80 s	Corresponds to an ideal 79.6 s 2-lap stint, matching top FSG vehicles.
Average Track Speed	83.19 km/h	High average cornering speed maintained through aerodynamic grip.
Top Straightline Speed	100.86 km/h	Reached at the end of the 75 m main straight under 80 kW ceiling.
Max Lateral Acceleration (ay)	3.59 g	Extreme grip enabled by Cl = 3.6 downforce and soft TTC compound.
Max Longitudinal Accel (ax)	+1.83 g	Traction limited at low speeds, power-limited at high speeds.
Max Braking Deceleration	-1.78 g	Four-wheel hydraulic + regenerative combined braking envelope.

6. Telemetry & Acceleration Diagrams

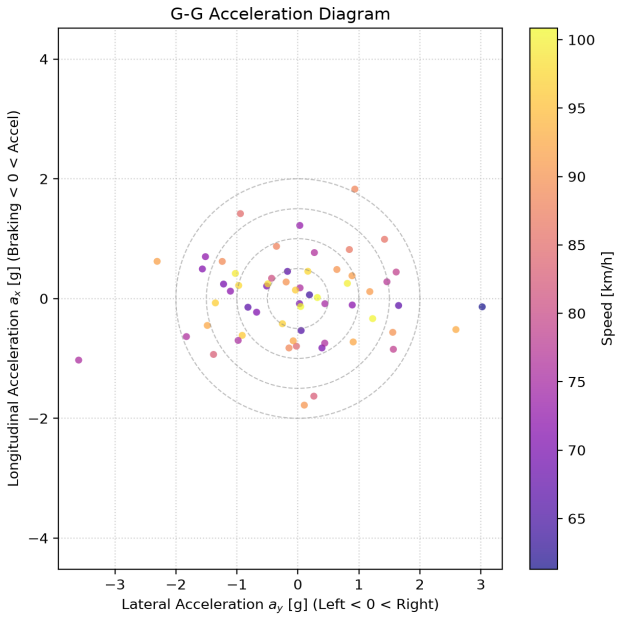


Figure 1: G-G Acceleration Diagram (ax vs. ay friction envelope)

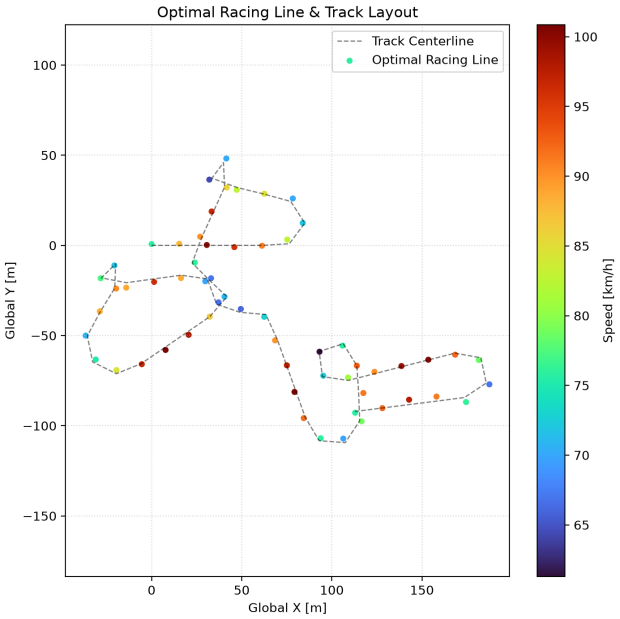


Figure 2: Optimal Racing Line & Track Centerline

7. Comprehensive Telemetry Channels

The figure below illustrates the synchronized telemetry traces along track arc-length *s*, including speed, throttle/brake pedal inputs, road wheel steering angle, motor power demand (kW), and battery SoC (%):

